import spacy from spacy import displacy from transformers import pipeline, AutoTokenizer, AutoMode1ForTokenC1assification import torch import pandas as pd

```
!        -m spacy download en_core_web_sm spacy.
    load ( " en_core_web_sm" )
```

cting en-core-web-sm==3.8.0

wnloading htt s: ithub.com ex losion s ac -models releases download en core web sm-3.8.Ø en core web sm-3.8.e-3-none-an

12.8/12.8 MB 86.3 MB/s eta

wnload and installation successful can now load he package via spacy.load('

estart to reload dependencies are in a Jupyter or Colab notebook, you may need to restart Python in to load all the package' s dependencies. You can do this by selecting the kernel' or ' Restart runtime' option.

python nlp
=

v'
You
A If
you
order
'Restart

```
sentences
    "Apple announced a new iPhone in California. "
    "Lionel Messi scored two goals for Argentina. " ,
    "The Prime Minister of India visited Hyderabad. " ,
    "Microsoft released new Al tools in 2026. " ,
    "The Olympics will be held in Paris. "
```

```
for sent in sentences:
    doc = nlp(sent) for
    ent in doc.ents:
        print(f"Sentence: {sent}\nEntity: {ent.text}\tLabe1: {ent . label_} " )
```

Sentence: Apple announced a new iPhone in California. Entity: Apple
Label: ORG

Sentence: Apple announced a new iPhone in California.
Entity: iPhone Label: ORG

Sentence: Apple announced a new iPhone in California.
Entity: California Label: GPE
Sentence: Lionel Messi scored two goals for Argentina.
Entity: Lionel Messi Label: PERSON
Sentence: Lionel Messi scored two goals for Argentina.
Entity: two Label :CARDINAL
Sentence: Lionel Messi scored two goals for Argentina.
Entity: Argentina Label: GPE
Sentence: The Prime Minister of India visited Hyderabad.
Entity: India Label: GPE
Sentence: The Prime Minister of India visited Hyderabad.
Entity: Hyderabad Label: ORG
Sentence: Microsoft released new Al tools in 2026.
Entity: Microsoft Label: ORG
Sentence: Microsoft released new Al tools in 2026.
Entity: Al Label: GPE

Sentence: Microsoft released new Al tools in 2026.
Entity: 2026 Label: DATE

Sentence: The Olympics will be held in Paris.

Entity: Olympics Label: EVENT

Sentence: The Olympics will be held in Paris.

Entity: Paris Label: GPE

```
results = for sent in
sentences: doc =
nlp(sent) for ent in
doc.ents:
```



```
results. append([sent, ent.text, ent . label_] )
df_spacy = pd. DataFrame(resu1ts, "Sentence", "Entity " , " Label " ] ) df_spacy
```

	Sentence	Entity	Label
0	Apple announced a new iPhone in California.	Apple	ORG
1	Apple announced a new iPhone in California.	iPhone	ORG
2	Apple announced a new iPhone in California.	California	
3	Lionel Messi scored two goals for Argentina.	Lionel Messi	PERSON
4	Lionel Messi scored two goals for Argentina.	two	CARDINAL
5	Lionel Messi scored two goals for Argentina.	Argentina	GPE
6	The Prime Minister of India visited Hyderabad.	India	
7	The Prime Minister of India visited Hyderabad.	Hyderabad	ORG
8	Microsoft released new Al tools in 2026.	Microsoft	ORG
9	Microsoft released new Al tools in 2026.		
10	Microsoft released new Al tools in 2026.	2026	DATE
11	The Olympics will be held in Paris.	Olympics	EVENT
12	The Olympics will be held in Paris.	Paris	GPE

Next steps: Generate code with df_spacy

```
displacy. render(doc, style="ent" , jupyter=True)
```

The Olympics EVENT will be held in Paris GPE

```
tokenizer = AutoTokenizer.from_pretrained( "ds1im/bert-base-NER") model =
AutoMode1ForTokenC1assification "ds1im/bert-base-NER") ner_pipeline =
pipeline( "ner" , model=model, tokenizer=tokenizer)
```

/usr/10ca1/1ib/python3.12/ dist-packages/ UserWarning:
The secret A HF_TOKEN* does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https: //huggingface.co/settings/tokens), set it You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets. warnings .
warn(

config.json: 100% 829/829 14.9kB/s]

tokenizer_config.json: 100% 59.0/59.0 1.23kB/s]

vocab. txt: 213k/? 1.59MB/s]

added_tokens.json: 100% 2.00/2.00 41.6B/s]

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING : huggingface_hub . utils . _http : Warning : You are sending unauthenticated requests to the HF Hub. Please set a
HF_TOKEN to en special_tokens_map.json: 100%      112/112   1.84kB/s] model.safetensors: 100%      433M/433M          75.5MB/s]
Loading weights: 100%                                                    199/199                585.88it/s, Materializing param=classifier.weight]
BertForTokenC1assification LOAD REPORT from: ds1im/bert-base-NER
Key                                  Status

bert . pooler. dense. bias UNEXPECTED I
bert. pooler.dense.weight I UNEXPECTED I
Notes :
- UNEXPECTED        : can be ignored when loading from different task/ architecture; not ok if you expect identical arch.
```

```
for sent in sentences :
    outputs      ner_pipeline(sent)
    print(f"Sentence: {sent}" ) for
    out in outputs:
        print(f"Word: {out [ 'word '            {out[ 'entity' ]}\tScore: {out [ 'score' ] : .2f}") print()
Sentence: Apple announced a new      iPhone in
                                             California.
Word: Apple        Entity : B.ORG     Score : 1.00
Word: iPhone       Entity : B-MISC    Score:  0.97
Word: California          Entity:B- LOC    Score: 1. ee
Sentence: Lionel Messi scored two goals for Argentina.
Word: Lionel  Entity :  B-PER   Score:  1.00
Word: Me      Entity :  I-PER   Score:  1.00
Word: ##ssi   Entity :  I-PER   Score:  1.00
Word: Argentina Entity:  B- LOC   Score   1.00
                                  :
Sentence: The Prime Minister of India visited Hyderabad.
Word: India    Entity: B-LOC  Score: 1.00
Word: Hyderabad Entity: B-LOC      Score : 1.00
Sentence: Microsoft released new Al tools in 2026.
Word: Microsoft Entity: B.ORG   Score:  1.00

Word: Al       Entity: B-MISC  Score : 0.94

Sentence: The Olympics will be      held in  Paris.

Word: Olympics Entity: B-MISC      Score:  0.99

Word: Paris    Entity: B-LOC   Score : 1.00
```

```
clean results for sent
in sentences:
    outputs = ner_pipeline(sent)  for
    out in outputs:
        word = out[ 'word' ] .                    )  # remove subword markers
        clean_ results . append([sent, word, out [ 'entity' ] , round(out[ ' score ' ] , 2) ] )
        pd .                                  "Sentence", "Entity" , " Label", "Confidence Score"] )
df hf
```

	Sentence	Entity	Label	Confidence Score
o	Apple announced a new iPhone in California.	Apple	B.ORG	1.00
1	Apple announced a new iPhone in California.	iPhone	B-MISC	0.97
2	Apple announced a new iPhone in California.	California	B-LOC	1.00
3	Lionel Messi scored two goals for Argentina.	Lionel	B-PER	1 00
4	Lionel Messi scored two goals for Argentina.	Me	I-PER	1 00
5	Lionel Messi scored two goals for Argentina.	ssi	I-PER	1 00

6	Lionel Messi scored two goals for Argentina.	Argentina	B-LOC	1.00
7	The Prime Minister of India visited Hyderabad.	India	B-LOC	
8	The Prime Minister of India visited Hyderabad.	Hyderabad	B-LOC	1 00
9	Microsoft released new Al tools in 2026.	Microsoft	B.ORG	1 00
10	Microsoft released new Al tools in 2026.		B-MISC	0.94
11	The Olympics will be held in Paris.	Olympics	B-MISC	0.99
12	The Olympics will be held in Paris.	Paris	B-LOC	1.00

Next steps: Generate code with df_hf

```
comparison = pd . DataFrame({
    " Feature" : ["Model Type" , "Speed" , "Accuracy", "Context Handling", "Confidence Score" ] ,
    "spaCy" • ["Rule-based + statistical", "Fast", "Good but limited" , "Weak in long context" , "Not provided"],
    "Hugging Face" : ["Transformer (BERT)" , "Slower", "Higher F 1" , "Strong contextual", "Yes"]
comparison
```

	Feature	spaCy	Hugging Face
0	Model Type	Rule-based + statistical	Transformer (BERT)
1	Speed	Fast	Slower
2	Accuracy	Good but limited	Higher F1
3	Context Handling	Weak in long context	Strong contextual
4	Confidence Score	Not provided	Yes

Next steps: Generate code with comparison New interactive sheet

```
from datasets import load_dataset

# CoNLL-2003
# The 'datasets' library in its current version no longer supports loading datasets directly from scripts like 'conll2003.py'.
# Please consider alternative ways to load this dataset, such as specific revisions or direct file downloads, or use an alterna
# conll = load_dataset("conll2003")

# WNUT-17
# The 'datasets' library in its current version no longer supports loading datasets directly from scripts like 'wnut_17.py'.
# Please consider alternative ways to load this dataset, such as specific revisions or direct file downloads, or use an alterna
# wnut = load_dataset("wnut_17")
```